

MATH 180 Graph Theory — Discussion 8

Counting labeled trees by degree sequence (Ch. 8.2)

Quick Recall (Ch. 8.1–8.2)

Spanning trees & notation. For a graph G , let $T(G)$ denote the number of spanning trees of G . Throughout, we assume the vertex set is $[n] = \{1, 2, \dots, n\}$.

Cayley's formula (Thm 8.1.1). For each $n \geq 2$, the number $T(K_n)$, i.e. the number of trees on given n vertices, equals n^{n-2} .

Handshake lemma in trees. Any tree on n vertices has exactly $n - 1$ edges, hence

$$\sum_{i=1}^n d_i = 2(n-1),$$

where d_i is the degree of vertex i .

Multinomial coefficient. The number of ways to partition $[n]$ into ordered groups of sizes $k_1, \dots, k_r$ (with $k_1 + \dots + k_r = n$) is

$$\binom{n}{k_1, k_2, \dots, k_r} = \frac{n!}{k_1! k_2! \dots k_r!}.$$

Proposition 8.2.1 (score-counting). Let $d_1, d_2, \dots, d_n$ be positive integers summing up to $2n - 2$. Then the number of spanning trees of the graph K_n in which the vertex i has degree exactly d_i for all $i = 1, 2, \dots, n$ equals

$$\frac{(n-2)!}{(d_1-1)!(d_2-1)! \dots (d_n-1)!}.$$

Proof idea. Induction on n : pick a leaf, remove it, and use a bijection onto trees of K_{n-1} with the decremented degree sequence.

Mini-example ($n = 4$, score $(3, 1, 1, 1)$). By the proposition,

$$\frac{2!}{2!0!0!0!} = 1$$

spanning tree of K_4 has vertex 1 of degree 3 – the star centered at 1.

Practice: Textbook Exercise 8.2.1* — trees with degrees in $\{1, 3\}$, part (a)

(a)* Find the number of trees (on a given vertex set $[n]$) in which all vertices have degree 1 or 3.

Hints

Suggested steps. (1) Let k be the number of degree-3 vertices, so $n - k$ vertices are leaves. Use the handshake lemma $\sum d_i = 2(n - 1)$ to solve for k in terms of n . (2) Observe what parity of n is forced. (3) Choose which vertices are degree-3 in $\binom{n}{k}$ ways; for each such choice, the degree sequence is fixed, so apply Proposition 8.2.1.

(b)* What if we allow degrees 1, 2, or 3?

 Hints

Suggested steps. (1) Let k be the number of degree-3 vertices and ℓ the number of degree-2 vertices; the remaining $n - k - \ell$ vertices are leaves. Use the handshake lemma to express ℓ in terms of k and n . (2) Determine which values of k are admissible (need $\ell \geq 0$ and $k \geq 0$). (3) For each admissible k : count the number of ways to choose which vertices play each role (multinomial coefficient), then apply Proposition 8.2.1. Sum over k .